LV ELECTRICAL

Calculation report

Villa Complex 415V LV Network · Plot 123-456, VILLA, UAE

Doc no.		**Rev**	
Date	15 October 2026		
Prepared		**Checked**	
Approved			

RESULTS SUMMARY

Study	Pass	Warning	Fail
Transformer and standby generator sizing	2	0	**0**
Cable and breaker sizing	36	0	**0**
Load flow and voltage drop study	22	0	**0**
Short circuit study	36	0	**0**
Earth fault loop impedance and disconnection	36	0	**0**
Protection discrimination study	0	31	**0**
Power factor correction	0	1	**0**

Needs attention (32)

INC-GF (500 A)	**WARNING** — Protection discrimination study: Downstream: GF-LTG (80 A) · In ratio: 6.25 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial
INC-GF (500 A)	**WARNING** — Protection discrimination study: Downstream: GF-SKT (40 A) · In ratio: 12.5 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial
INC-GF (500 A)	**WARNING** — Protection discrimination study: Downstream: GF-HVAC (250 A) · In ratio: 2 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial
INC-GF (500 A)	**WARNING** — Protection discrimination study: Downstream: INC-DBGF1 (100 A) · In ratio: 5 · Fault at downstream (kA): 21.8 · Selectivity limit (kA): 4 · Short circuit: Partial
INC-FF (200 A)	**WARNING** — Protection discrimination study: Downstream: FF-OFF (63 A) · In ratio: 3.17 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial
INC-FF (200 A)	**WARNING** — Protection discrimination study: Downstream: FF-IT (40 A) · In ratio: 5 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial
INC-FF (200 A)	**WARNING** — Protection discrimination study: Downstream: FF-LAB (80 A) · In ratio: 2.5 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial
INC-FF (200 A)	**WARNING** — Protection discrimination study: Downstream: FF-LTG (25 A) · In ratio: 8 · Fault at downstream (kA): 14.3 · Selectivity limit (kA): 1.6 · Short circuit: Partial
INC-MCC (315 A)	**WARNING** — Protection discrimination study: Downstream: MCC-WP (160 A) · In ratio: 1.97 · Fault at downstream (kA): 19 · Selectivity limit (kA): 2.5 · Short circuit: Partial
INC-MCC (315 A)	**WARNING** — Protection discrimination study: Downstream: MCC-FP (125 A) · In ratio: 2.52 · Fault at downstream (kA): 19 · Selectivity limit (kA): 2.5 · Short circuit: Partial

…and 22 more — see the sections below.

DESIGN BASIS

Parameter	Value
System	415 V, 3-phase + N, 50 Hz, TN-S (as calculated)
Design ambient temperature	45 °C
Voltage drop limit (source to load)	4 %
Cable	Copper; XLPE/PVC/SWA; rated for each circuit's installation method (IEC 60364-5-52)
Standards	IEC 60364-4-41, 4-43, 5-52, 5-54; IEC 60909-based simplified fault levels (preliminary); IEC 60898 / 60947-2 (breakers)
Sizing targets	Transformer ≤ 80% with 20% growth; generator ≤ 80%; PF target 0.95

1. LOAD ASSESSMENT

Input

Loads	Connected loads on the feeders and DB load schedules
Demand factors	Per load / circuit as entered

Design criteria

Future growth allowance	20 % (app default)
Transformer design loading	80 %

Method

- Demand = connected load × demand factor for each load, summed up the network (kW and kvar separately); maximum demand kVA = $\sqrt{(kW^2 + kvar^2)}$.
- Required capacity = maximum demand × (1 + growth) ÷ design loading, from the transformer sizing (no further diversity applied).

Result

Connected load	514 kW
Maximum demand	451 kW
Demand ÷ connected	0.88

Load at each board

Board	Connected (kW)	Demand (kW)	Demand ÷ connected	Demand (kVA)	PF	Current (A)
MDB-1	514	451.1	0.88	512.1	0.88	712
SMDB-GF	231	203.3	0.88	233.7	0.87	325
DB-GF1	38	38	1	42.2	0.9	59
SMDB-FF	109	87	0.8	97.4	0.89	135
MCC-1	130	130	1	151.2	0.86	210

Design load of each main board

Main board	Maximum demand (kVA)	Future growth	Required capacity (kVA)
MDB-1	473	20 %	710

2. TRANSFORMER AND STANDBY GENERATOR SIZING

Overall status: **PASS** (2 pass · 0 warning · 0 fail)

Design criteria

Future growth	20 %
Transformer design loading	≤ 80 %
Generator design loading	≤ 80 %

Method

- Transformer per main board: maximum demand (after the planned power factor correction) × (1 + 20 % growth) ÷ 80 % design loading, next DEWA standard size (500 / 1000 / 1500 kVA).
- For that size: full-load current and main breaker, LV fault level (typical IEC 60076-5 impedance, infinite MV source), voltage regulation at the design demand.
- Bus couplers: with one transformer out, the other carries both boards up to 100 % of its rating. Duty / standby: two transformers, each for the whole load.
- Standby generator from the boards on it (share of each), plus circuits marked essential. The set must carry the running kVA and the running kW at 80 % loading: rating ≥ max(kVA ÷ 0.8, kW ÷ (0.8 × 0.8)), with rated kW = 0.8 × rated kVA unless the Generators list gives the set's kW; or larger if the largest motor, started last, would dip the voltage over 15 % (X′d 25 %).

Result

Items checked	2 pass · 0 warning · 0 fail
MDB-1 transformer	1000 kVA

Standby generator	630 kVA / 504 kW

Transformers

Main board	Demand (kVA)	PF	Design (kVA)	Recommended	Installed · loading	FLC · main breaker	LV fault (kA) · lowest Icu	Regulation	Result
MDB-1	473	0.95	710	1000 kVA	1000 kVA · 47 %	1391 A · 1600 A	29.2 · 36	1.1 %	PASS

Standby generator — loads

Board / circuit	Share	Demand (kW)	Reactive (kvar)
MCC-FP Fire pump (essential circuit)	100 %	55	—

Standby generator — size

Item	Value
Running demand	55 kW · 64 kVA
For the running load	86 kVA — running; the motor start sets the size
Largest motor start	MCC-FP: 384 kVA starting with 0 kVA already running
For the motor start	544 kVA (dip 13.2 % on the recommended set)
Recommended	630 kVA / 504 kW at 0.8 PF
Result	PASS
With a soft starter on the largest motor	300 kVA
Full-load current · ATS / breaker	876 A · 1000 A
Installed	—

Verification — Capacity verification

Equipment	Calculated demand (kVA)	Required capacity (kVA)	Selected	Installed (kVA)	Loading	Spare (kVA)	Status
Transformer MDB-1	473	710	1000 kVA	1000	47 %	527	PASS
Standby generator	64	544	630 kVA	—	—	—	*NOT CHECKED*

3. CABLE AND BREAKER SIZING

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

Input

Cable	Cu / XLPE multicore, rated for each circuit's installation method (IEC 60364-5-52)
Ambient temperature	45 °C
Ground (buried cables)	20 °C, soil 2.5 K·m/W (IEC reference where not set)
Grouping	From the cable tray route (method E), else the circuit's own parallel runs

Design criteria

Overload protection	Ib ≤ In ≤ Iz (IEC 60364-4-43)
Voltage drop limit	4 %
Breaking capacity	Icu ≥ Ik″ at the breaker

Method

- Cu/XLPE/SWA multicore cable, rated for each circuit's IEC 60364-5-52 installation method (E free air / tray unless set); derated for air ambient (methods A–E) or ground temperature and soil thermal resistivity (D1 / D2, reference 20 °C and 2.5 K·m/W), and for grouping (the cable tray route for method E, else the circuit's own parallel runs: bunched for A–C, in ducts / in the ground for D1 / D2).
- Overload protection per IEC 60364-4-43: Ib ≤ In ≤ Iz. Voltage drop from the main board to the load within the limit. Breaker Icu ≥ the fault at its terminals.

Result

Circuits checked	36 pass · 0 warning · 0 fail
Ambient	45 °C

Cables and breakers

Circuit	From	To	Cable	Installation	Derating	Ib (A)	Breaker In (A)	Iz (A)	Ib ≤ In ≤ Iz	ΔV total (%)	Result
INC-GF	MDB-1	SMDB-GF	4C × 300 mm² XLPE/SWA + 150 CPC	E	0.87 · 1.00	326.8	500 MCCB	525	PASS	0.46	PASS
INC-FF	MDB-1	SMDB-FF	4C × 95 mm² XLPE/SWA + 50 CPC	E	0.87 · 1.00	135.5	200 MCCB	266	PASS	0.75	PASS
INC-MCC	MDB-1	MCC-1	4C × 150 mm² XLPE/SWA + 95 CPC	E	0.87 · 1.00	210.3	315 MCCB	342	PASS	0.58	PASS
DB-EV	MDB-1	EV charging – parking	4C × 10 mm² XLPE/SWA + 10 CPC	E	0.87 · 1.00	43.7	63 C	70	PASS	2.38	PASS
DB-PV	MDB-1	Solar PV inverter	4C × 25 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	69.6	100 MCCB	120	PASS	0.76	PASS

Circuit	From	To	Cable	Installation	Derating	Ib (A)	Breaker In (A)	Iz (A)	Ib ≤ In ≤ Iz	ΔV total (%)	Result
GF-LTG	SMDB-GF	Lighting	4C × 16 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	62.6	80 MCCB	91	PASS	1.47	PASS
GF-SKT	SMDB-GF	Sockets	4C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	27.5	40 C	41	PASS	2.37	PASS
GF-HVAC	SMDB-GF	HVAC	4C × 95 mm² XLPE/SWA + 50 CPC	E	0.87 · 1.00	176.8	250 MCCB	266	PASS	1.16	PASS
FF-OFF	SMDB-FF	Office	4C × 10 mm² XLPE/SWA + 10 CPC	E	0.87 · 1.00	43.3	63 C	70	PASS	1.66	PASS
FF-IT	SMDB-FF	IT room	4C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	29.3	40 C	41	PASS	2.04	PASS
FF-LAB	SMDB-FF	Lab	4C × 16 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	57.3	80 MCCB	91	PASS	1.62	PASS
FF-LTG	SMDB-FF	Corridor lighting	2C × 6 mm² XLPE/SWA + 6 CPC	E	0.87 · 1.00	18.5	25 C	52	PASS	2.57	PASS
MCC-WP	MCC-1	Water pump	4C × 50 mm² XLPE/SWA + 25 CPC	E	0.87 · 1.00	121.3	160 MCCB	174	PASS	1.03	PASS
MCC-FP	MCC-1	Fire pump	4C × 35 mm² FR + 16 CPC	E	0.87 · 1.00	89	125 MCCB	146	PASS	1.12	PASS
INC-DBGF1	SMDB-GF	DB-GF1	4C × 35 mm² XLPE/SWA + 16 CPC	E	0.87 · 1.00	60.4	100 MCCB	146	PASS	0.92	PASS
DB-GF1-Y1	DB-GF1	Majlis lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	4.6	6 C	31	PASS	1.48	PASS
DB-GF1-R1	DB-GF1	Living lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	5.6	10 C	31	PASS	1.67	PASS
DB-GF1-Y2	DB-GF1	Kitchen lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	3.9	6 C	31	PASS	1.49	PASS
DB-GF1-B6	DB-GF1	Majlis sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	7	10 C	41	PASS	1.44	PASS
DB-GF1-Y5	DB-GF1	Living sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	9.3	16 C	41	PASS	1.7	PASS
DB-GF1-B7	DB-GF1	Kitchen sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	5.8	10 C	41	PASS	1.45	PASS
DB-GF1-R3	DB-GF1	Cooker	2C × 6 mm² XLPE/SWA + 6 CPC	E	0.87 · 1.00	27.8	40 C	52	PASS	2.79	PASS
DB-GF1-B3	DB-GF1	Kitchen water heater	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	13.9	20 C	41	PASS	2.32	PASS
DB-GF1-R6	DB-GF1	Maid room	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	5.1	6 C	41	PASS	1.51	PASS
DB-GF1-B4	DB-GF1	Majlis split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	1.79	PASS
DB-GF1-Y4	DB-GF1	Living split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	1.89	PASS
DB-GF1-B5	DB-GF1	Dining split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	1.99	PASS
DB-GF1-R5	DB-GF1	Guest bedroom	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	6.9	10 C	41	PASS	1.67	PASS
DB-GF1-Y8	DB-GF1	Guest bathroom	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	1.2	6 C	41	PASS	1.05	PASS
DB-GF1-Y3	DB-GF1	Guest bath water heater	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	13.9	20 C	41	PASS	2.49	PASS
DB-GF1-R4	DB-GF1	Guest bedroom split A/C	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	11.6	16 C	41	PASS	2.18	PASS

Circuit	From	To	Cable	Installation	Derating	Ib (A)	Breaker In (A)	Iz (A)	Ib ≤ In ≤ Iz	ΔV total (%)	Result
DB-GF1-B2	DB-GF1	Dining lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	2.8	6 C	31	PASS	1.33	PASS
DB-GF1-Y7	DB-GF1	Dining sockets	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	4.6	6 C	41	PASS	1.35	PASS
DB-GF1-B1	DB-GF1	External lighting	2C × 2.5 mm² XLPE/SWA + 2.5 CPC	E	0.87 · 1.00	4.6	6 C	31	PASS	2.01	PASS
DB-GF1-R7	DB-GF1	Water pump	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	3.5	6 C	41	PASS	1.36	PASS
DB-GF1-Y6	DB-GF1	Laundry	2C × 4 mm² XLPE/SWA + 4 CPC	E	0.87 · 1.00	9.3	16 C	41	PASS	1.89	PASS

Verification — Circuits on each board

Board	Circuits	Pass / check / fail	Status
MDB-1	5	5 / 0 / 0	PASS
SMDB-GF	4	4 / 0 / 0	PASS
DB-GF1	21	21 / 0 / 0	PASS
SMDB-FF	4	4 / 0 / 0	PASS
MCC-1	2	2 / 0 / 0	PASS

4. LOAD FLOW AND VOLTAGE DROP STUDY

Overall status: **PASS** (22 pass · 0 warning · 0 fail)

Input

Supply	MDB-1: transformer 1000 kVA, 5 % Z
Loads	Design loads × demand factors (load schedules and feeders)

Design criteria

Voltage drop limit (source to load)	4 %

Method

- Balanced 3-phase load flow from the design loads: demand = connected load × demand factor, summed up the network (kW and kVAr separately).
- Voltage drop per cable: √3 · I · L · (R cosφ + X sinφ) for 3-phase, 2 · I · L · (…) for single-phase; busbar voltage = nominal minus the drops of the incomers above it. Limit 4 % from the main board to the load.
- Transformer regulation is not included (busbar voltages are measured from the main LV busbar).

- Motors: drop while starting = upstream drop + cable drop × the starter's starting-current multiple; limit 10 %.

Result

Total demand of the scope	451 kW · 512 kVA
Lowest busbar voltage	99.08 %
Circuits checked	15 pass · 0 warning · 0 fail

Busbars

Board	Connected (kW)	Demand (kW)	Demand (kVA)	PF	Current (A)	Rating (A)	Loading	Voltage (V)	% of nominal
MDB-1	514	451.1	512.1	0.88	712	1600	45 %	415	100
SMDB-GF	231	203.3	233.7	0.87	325	630	52 %	413.1	99.54
DB-GF1	38	38	42.2	0.9	59	100	59 %	411.2	99.08
SMDB-FF	109	87	97.4	0.89	135	250	54 %	411.9	99.25
MCC-1	130	130	151.2	0.86	210	400	53 %	412.6	99.42

Feeders

Circuit	From	To	Cable	Length (m)	Ib (A)	PF	ΔV cable (%)	ΔV total (%)	Result
INC-GF	MDB-1	SMDB-GF	4C × 300 mm²	35	326.8	0.87	0.46	0.46	PASS
INC-FF	MDB-1	SMDB-FF	4C × 95 mm²	55	135.5	0.89	0.75	0.75	PASS
INC-MCC	MDB-1	MCC-1	4C × 150 mm²	40	210.3	0.86	0.58	0.58	PASS
DB-EV	MDB-1	EV charging – parking	4C × 10 mm²	60	43.7	0.98	2.38	2.38	PASS
DB-PV	MDB-1	Solar PV inverter	4C × 25 mm²	30	69.6	1	0.76	0.76	PASS
GF-LTG	SMDB-GF	Lighting	4C × 16 mm²	30	62.6	0.9	1	1.47	PASS
GF-SKT	SMDB-GF	Sockets	4C × 4 mm²	35	27.5	0.85	1.91	2.37	PASS
GF-HVAC	SMDB-GF	HVAC	4C × 95 mm²	40	176.8	0.85	0.7	1.16	PASS
FF-OFF	SMDB-FF	Office	4C × 10 mm²	25	43.3	0.9	0.91	1.66	PASS
FF-IT	SMDB-FF	IT room	4C × 4 mm²	20	29.3	0.95	1.29	2.04	PASS
FF-LAB	SMDB-FF	Lab	4C × 16 mm²	30	57.3	0.85	0.87	1.62	PASS
FF-LTG	SMDB-FF	Corridor lighting	2C × 6 mm²	35	18.5	0.9	1.82	2.57	PASS

Circuit	From	To	Cable	Length (m)	Ib (A)	PF	ΔV cable (%)	ΔV total (%)	Result
MCC-WP	MCC-1	Water pump	4C × 50 mm²	20	121.3	0.86	0.45	1.03	PASS
MCC-FP	MCC-1	Fire pump	4C × 35 mm²	25	89	0.86	0.54	1.12	PASS
INC-DBGF1	SMDB-GF	DB-GF1	4C × 35 mm²	30	60.4	0.9	0.45	0.92	PASS

Motor starting — limit 10 % source to motor

Circuit	From	Motor	Running ΔV total (%)	Starting ΔV total (%)	Limit (%)	Result
MCC-WP	MCC-1	Motor (DOL)	1.03	3.3	10	PASS
MCC-FP	MCC-1	Fire pump (DOL)	1.12	3.8	10	PASS

Verification — Voltage drop — worst circuit on each board

Board	Circuit	Calculated ΔV total (%)	Allowable (%)	Margin (%)	Status
MDB-1	DB-EV	2.38	4	1.62	PASS
SMDB-GF	GF-SKT	2.37	4	1.63	PASS
SMDB-FF	FF-LTG	2.57	4	1.43	PASS
MCC-1	MCC-FP	1.12	4	2.88	PASS

5. SHORT CIRCUIT STUDY

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

Input

Supply	MDB-1: transformer 1000 kVA, 5 % Z
System	415 V, 50 Hz

Design criteria

Voltage factor	c max = 1.05 (maximum fault, cable R at 20 °C)
Breaking capacity	Breaker Icu ≥ prospective fault at its terminals (Ik″)

Method

- Maximum prospective symmetrical 3-phase fault current Ik″, simplified method based on IEC 60909-0 (preliminary — not a full IEC 60909 study): voltage factor c max = 1.05, cable R at 20 °C, upstream MV network infinite, transformer correction factor KT not applied.
- Source impedance from the transformer rating and %Z (split by its X/R ratio); cable R at 20 °C and X per km; parallel runs divide the impedance.
- Breaking capacity check: the fault at the breaker terminals (supply busbar) must not exceed the breaker Icu. Fault at the cable end: 3-phase for 3/4-core circuits, line-to-neutral for 2-core.
- Motor contribution and generation are neglected.

Result

Highest busbar fault	29.2 kA
Breakers checked	36 pass · 0 warning · 0 fail

Fault levels at the busbars

Board	Type	Fed from	Busbar rating (A)	Ik″ max (kA)
MDB-1	MDB	Transformer 1000 kVA, 5 % Z	1600	29.2
SMDB-GF	SMDB	MDB-1	630	21.8
DB-GF1	DB	SMDB-GF	100	10.7
SMDB-FF	SMDB	MDB-1	250	14.3
MCC-1	MCC	MDB-1	400	19

Breaker breaking capacity

Circuit	Board	To	Breaker	Ik″ at breaker (kA)	Icu (kA)	Margin	Result
INC-GF	MDB-1	SMDB-GF	500 A MCCB	29.2	36	23 %	PASS
INC-FF	MDB-1	SMDB-FF	200 A MCCB	29.2	36	23 %	PASS
INC-MCC	MDB-1	MCC-1	315 A MCCB	29.2	36	23 %	PASS
DB-EV	MDB-1	EV charging – parking	63 A C	29.2	36	23 %	PASS
DB-PV	MDB-1	Solar PV inverter	100 A MCCB	29.2	36	23 %	PASS
GF-LTG	SMDB-GF	Lighting	80 A MCCB	21.8	25	15 %	PASS
GF-SKT	SMDB-GF	Sockets	40 A C	21.8	25	15 %	PASS
GF-HVAC	SMDB-GF	HVAC	250 A MCCB	21.8	25	15 %	PASS
FF-OFF	SMDB-FF	Office	63 A C	14.3	25	75 %	PASS

Circuit	Board	To	Breaker	Ik″ at breaker (kA)	Icu (kA)	Margin	Result
FF-IT	SMDB-FF	IT room	40 A C	14.3	25	75 %	PASS
FF-LAB	SMDB-FF	Lab	80 A MCCB	14.3	25	75 %	PASS
FF-LTG	SMDB-FF	Corridor lighting	25 A C	14.3	25	75 %	PASS
MCC-WP	MCC-1	Water pump	160 A MCCB	19	25	32 %	PASS
MCC-FP	MCC-1	Fire pump	125 A MCCB	19	25	32 %	PASS
INC-DBGF1	SMDB-GF	DB-GF1	100 A MCCB	21.8	25	15 %	PASS
DB-GF1-Y1	DB-GF1	Majlis lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-R1	DB-GF1	Living lighting	10 A C	10.7	15	41 %	PASS
DB-GF1-Y2	DB-GF1	Kitchen lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-B6	DB-GF1	Majlis sockets	10 A C	10.7	15	41 %	PASS
DB-GF1-Y5	DB-GF1	Living sockets	16 A C	10.7	15	41 %	PASS
DB-GF1-B7	DB-GF1	Kitchen sockets	10 A C	10.7	15	41 %	PASS
DB-GF1-R3	DB-GF1	Cooker	40 A C	10.7	15	41 %	PASS
DB-GF1-B3	DB-GF1	Kitchen water heater	20 A C	10.7	15	41 %	PASS
DB-GF1-R6	DB-GF1	Maid room	6 A C	10.7	15	41 %	PASS
DB-GF1-B4	DB-GF1	Majlis split A/C	16 A C	10.7	15	41 %	PASS
DB-GF1-Y4	DB-GF1	Living split A/C	16 A C	10.7	15	41 %	PASS
DB-GF1-B5	DB-GF1	Dining split A/C	16 A C	10.7	15	41 %	PASS
DB-GF1-R5	DB-GF1	Guest bedroom	10 A C	10.7	15	41 %	PASS
DB-GF1-Y8	DB-GF1	Guest bathroom	6 A C	10.7	15	41 %	PASS
DB-GF1-Y3	DB-GF1	Guest bath water heater	20 A C	10.7	15	41 %	PASS
DB-GF1-R4	DB-GF1	Guest bedroom split A/C	16 A C	10.7	15	41 %	PASS
DB-GF1-B2	DB-GF1	Dining lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-Y7	DB-GF1	Dining sockets	6 A C	10.7	15	41 %	PASS
DB-GF1-B1	DB-GF1	External lighting	6 A C	10.7	15	41 %	PASS
DB-GF1-R7	DB-GF1	Water pump	6 A C	10.7	15	41 %	PASS

Circuit	Board	To	Breaker	Ik″ at breaker (kA)	Icu (kA)	Margin	Result
DB-GF1-Y6	DB-GF1	Laundry	16 A C	10.7	15	41 %	PASS

Fault at the end of each cable

Circuit	From	To	Cable	Length (m)	Ik at cable end (kA)
INC-GF	MDB-1	SMDB-GF	4C × 300 mm²	35	21.81
INC-FF	MDB-1	SMDB-FF	4C × 95 mm²	55	14.3
INC-MCC	MDB-1	MCC-1	4C × 150 mm²	40	19.01
DB-EV	MDB-1	EV charging – parking	4C × 10 mm²	60	2.24
DB-PV	MDB-1	Solar PV inverter	4C × 25 mm²	30	9.7
GF-LTG	SMDB-GF	Lighting	4C × 16 mm²	30	6.2
GF-SKT	SMDB-GF	Sockets	4C × 4 mm²	35	1.52
GF-HVAC	SMDB-GF	HVAC	4C × 95 mm²	40	13.94
FF-OFF	SMDB-FF	Office	4C × 10 mm²	25	4.2
FF-IT	SMDB-FF	IT room	4C × 4 mm²	20	2.38
FF-LAB	SMDB-FF	Lab	4C × 16 mm²	30	5.11
FF-LTG	SMDB-FF	Corridor lighting	2C × 6 mm²	35	1.1
MCC-WP	MCC-1	Water pump	4C × 50 mm²	20	12.95
MCC-FP	MCC-1	Fire pump	4C × 35 mm²	25	10.53
INC-DBGF1	SMDB-GF	DB-GF1	4C × 35 mm²	30	10.66
DB-GF1-Y1	DB-GF1	Majlis lighting	2C × 2.5 mm²	18	0.88
DB-GF1-R1	DB-GF1	Living lighting	2C × 2.5 mm²	20	0.8
DB-GF1-Y2	DB-GF1	Kitchen lighting	2C × 2.5 mm²	22	0.73
DB-GF1-B6	DB-GF1	Majlis sockets	2C × 4 mm²	18	1.35
DB-GF1-Y5	DB-GF1	Living sockets	2C × 4 mm²	20	1.23
DB-GF1-B7	DB-GF1	Kitchen sockets	2C × 4 mm²	22	1.13
DB-GF1-R3	DB-GF1	Cooker	2C × 6 mm²	24	1.5

Circuit	From	To	Cable	Length (m)	Ik at cable end (kA)
DB-GF1-B3	DB-GF1	Kitchen water heater	2C × 4 mm²	24	1.04
DB-GF1-R6	DB-GF1	Maid room	2C × 4 mm²	28	0.9
DB-GF1-B4	DB-GF1	Majlis split A/C	2C × 4 mm²	18	1.35
DB-GF1-Y4	DB-GF1	Living split A/C	2C × 4 mm²	20	1.23
DB-GF1-B5	DB-GF1	Dining split A/C	2C × 4 mm²	22	1.13
DB-GF1-R5	DB-GF1	Guest bedroom	2C × 4 mm²	26	0.97
DB-GF1-Y8	DB-GF1	Guest bathroom	2C × 4 mm²	27	0.94
DB-GF1-Y3	DB-GF1	Guest bath water heater	2C × 4 mm²	27	0.94
DB-GF1-R4	DB-GF1	Guest bedroom split A/C	2C × 4 mm²	26	0.97
DB-GF1-B2	DB-GF1	Dining lighting	2C × 2.5 mm²	22	0.73
DB-GF1-Y7	DB-GF1	Dining sockets	2C × 4 mm²	22	1.13
DB-GF1-B1	DB-GF1	External lighting	2C × 2.5 mm²	35	0.47
DB-GF1-R7	DB-GF1	Water pump	2C × 4 mm²	30	0.85
DB-GF1-Y6	DB-GF1	Laundry	2C × 4 mm²	25	1

Verification — Breaking capacity at each busbar

Location	Ik″ busbar (kA)	Governing circuit	Ik″ at breaker (kA)	Icu (kA)	Margin (kA)	Breakers pass / check / fail	Status
MDB-1	29.2	INC-GF	29.2	36	6.8	5 / 0 / 0	PASS
SMDB-GF	21.8	GF-LTG	21.8	25	3.2	4 / 0 / 0	PASS
DB-GF1	10.7	DB-GF1-Y1	10.7	15	4.3	21 / 0 / 0	PASS
SMDB-FF	14.3	FF-OFF	14.3	25	10.7	4 / 0 / 0	PASS
MCC-1	19	MCC-WP	19	25	6	2 / 0 / 0	PASS

6. EARTH FAULT LOOP IMPEDANCE AND DISCONNECTION

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

Input

Earthing arrangement	TN-S
Supply loop (Ze)	From the source impedance to each board

Design criteria

Disconnection time	IEC 60364-4-41 Table 41.1: 0.4 s final circuits, 5 s distribution circuits
Fault current	Minimum, c = 0.95
Protective conductor	Adiabatic, k = 143

Method

- TN-S system. Earth fault loop impedance Zs = Ze (at the supply board) + (R phase + R cpc) of the circuit, at operating temperature; minimum fault current with c = 0.95.
- Protective conductor (adiabatic): S ≥ I·√t ÷ 143, t = 0.1 s when tripping instantaneously (not the device let-through energy), with I the current through the conductor checked. Parallel runs: each CPC carries an equal share of the end-of-circuit fault (identical runs bonded at both ends); a result that passes only on that assumption is marked "assumed sharing" (warning) because a fault within one run or unequal runs are not covered. The total fault current sets the disconnection check.
- Disconnection per IEC 60364-4-41 Table 41.1 (TN, U0 in the 230 V band): 0.4 s for final circuits up to 63 A with socket-outlets and up to 32 A supplying fixed equipment only; 5 s for distribution circuits and other final circuits. Circuits whose purpose is not set are treated as socket-outlet circuits. Instantaneous tripping when If ≥ Ia; below that, a 5 s circuit needs the breaker's time-current curve checked. Protective conductor checked by the adiabatic equation (k = 143).

Result

Circuits checked	36 pass · 0 warning · 0 fail

Earth fault loop

Circuit	Board	Breaker	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Result
INC-GF	MDB-1	500 A MCCB	150	0.0144	0.0455	15804	5000	5	< 0.1 s	34.9	PASS
INC-FF	MDB-1	200 A MCCB	50	0.0419	0.1138	5433	2000	5	< 0.1 s	12	PASS
INC-MCC	MDB-1	315 A MCCB	95	0.0204	0.0723	11150	3150	5	< 0.1 s	24.7	PASS
DB-EV	MDB-1	63 A C	10	0.2656	0.3613	857	630	0.4	< 0.1 s	1.9	PASS
DB-PV	MDB-1	100 A MCCB	16	0.0701	0.2276	3246	1000	5	< 0.1 s	7.2	PASS
GF-LTG	SMDB-GF	80 A MCCB	16	0.0932	0.2845	2442	800	5	< 0.1 s	5.4	PASS
GF-SKT	SMDB-GF	40 A C	4	0.3969	0.5691	573	400	0.4	< 0.1 s	1.3	PASS
GF-HVAC	SMDB-GF	250 A MCCB	50	0.0398	0.091	5724	2500	5	< 0.1 s	12.7	PASS

Circuit	Board	Breaker	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Result
FF-OFF	SMDB-FF	63 A C	10	0.1505	0.3613	1512	630	0.4	< 0.1 s	3.3	PASS
FF-IT	SMDB-FF	40 A C	4	0.2616	0.5691	870	400	0.4	< 0.1 s	1.9	PASS
FF-LAB	SMDB-FF	80 A MCCB	16	0.1237	0.2845	1840	800	5	< 0.1 s	4.1	PASS
FF-LTG	SMDB-FF	25 A C	6	0.2991	0.9105	761	250	0.4	< 0.1 s	1.7	PASS
MCC-WP	MCC-1	160 A MCCB	25	0.0455	0.1423	4997	1600	5	< 0.1 s	11.1	PASS
MCC-FP	MCC-1	125 A MCCB	16	0.0685	0.1821	3325	1250	5	< 0.1 s	7.4	PASS
INC-DBGF1	SMDB-GF	100 A MCCB	16	0.0709	0.2276	3209	1000	5	< 0.1 s	7.1	PASS
DB-GF1-Y1	DB-GF1	6 A C	2.5	0.3901	455.2407	584	1	0.4	< 0.1 s	1.3	PASS
DB-GF1-R1	DB-GF1	10 A C	2.5	0.4256	455.2407	535	1	0.4	< 0.1 s	1.2	PASS
DB-GF1-Y2	DB-GF1	6 A C	2.5	0.4612	455.2407	494	1	0.4	< 0.1 s	1.1	PASS
DB-GF1-B6	DB-GF1	10 A C	4	0.2693	1517.469	845	0	0.4	< 0.1 s	1.9	PASS
DB-GF1-Y5	DB-GF1	16 A C	4	0.2914	1517.469	781	0	0.4	< 0.1 s	1.7	PASS
DB-GF1-B7	DB-GF1	10 A C	4	0.3135	1517.469	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-R3	DB-GF1	40 A C	6	0.2476	1517.469	919	0	0.4	< 0.1 s	2	PASS
DB-GF1-B3	DB-GF1	20 A C	4	0.3356	1517.469	678	0	0.4	< 0.1 s	1.5	PASS
DB-GF1-R6	DB-GF1	6 A C	4	0.3798	1517.469	599	0	0.4	< 0.1 s	1.3	PASS
DB-GF1-B4	DB-GF1	16 A C	4	0.2693	1517.469	845	0	0.4	< 0.1 s	1.9	PASS
DB-GF1-Y4	DB-GF1	16 A C	4	0.2914	1517.469	781	0	0.4	< 0.1 s	1.7	PASS
DB-GF1-B5	DB-GF1	16 A C	4	0.3135	1517.469	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-R5	DB-GF1	10 A C	4	0.3577	1517.469	636	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-Y8	DB-GF1	6 A C	4	0.3687	1517.469	617	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-Y3	DB-GF1	20 A C	4	0.3687	1517.469	617	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-R4	DB-GF1	16 A C	4	0.3577	1517.469	636	0	0.4	< 0.1 s	1.4	PASS
DB-GF1-B2	DB-GF1	6 A C	2.5	0.4612	455.2407	494	1	0.4	< 0.1 s	1.1	PASS
DB-GF1-Y7	DB-GF1	6 A C	4	0.3135	1517.469	726	0	0.4	< 0.1 s	1.6	PASS
DB-GF1-B1	DB-GF1	6 A C	2.5	0.6923	455.2407	329	1	0.4	< 0.1 s	0.7	PASS

Circuit	Board	Breaker	CPC (mm²)	Zs (Ω)	Max Zs (Ω)	If (A)	Ia (A)	Required (s)	Disconnection	CPC min, each (mm²)	Result
DB-GF1-R7	DB-GF1	6 A C	4	0.4019	1517.469	566	0	0.4	< 0.1 s	1.3	PASS
DB-GF1-Y6	DB-GF1	16 A C	4	0.3466	1517.469	657	0	0.4	< 0.1 s	1.5	PASS

Verification — Earth fault loop on each board

Board	Circuits	Highest Zs ÷ max Zs	Pass / check / fail	Status
MDB-1	5	74 %	5 / 0 / 0	PASS
SMDB-GF	4	70 %	4 / 0 / 0	PASS
DB-GF1	21	0 %	21 / 0 / 0	PASS
SMDB-FF	4	46 %	4 / 0 / 0	PASS
MCC-1	2	38 %	2 / 0 / 0	PASS

7. PROTECTION DISCRIMINATION STUDY

Overall status: **WARNING** (0 pass · 31 warning · 0 fail)

Design criteria

Overload selectivity	In upstream ÷ In downstream ≥ 1.6
Short-circuit selectivity	Fault at the downstream breaker below the upstream magnetic threshold

Method

- Each breaker against the breaker above it. Overload (time) selectivity: In upstream ÷ In downstream ≥ 1.6.
- Short-circuit selectivity: total when the maximum fault at the downstream breaker stays below the upstream magnetic no-trip threshold; otherwise partial up to that current. Confirm with the manufacturer's selectivity tables.

Result

Pairs checked	**0 pass · 31 warning · 0 fail**

Selectivity

Upstream	Downstream	In ratio	Fault at downstream (kA)	Selectivity limit (kA)	Short circuit	Result
INC-GF (500 A)	GF-LTG (80 A)	6.25	21.8	4	Partial	**WARNING**

Upstream	Downstream	In ratio	Fault at downstream (kA)	Selectivity limit (kA)	Short circuit	Result
INC-GF (500 A)	GF-SKT (40 A)	12.5	21.8	4	Partial	**WARNING**
INC-GF (500 A)	GF-HVAC (250 A)	2	21.8	4	Partial	**WARNING**
INC-GF (500 A)	INC-DBGF1 (100 A)	5	21.8	4	Partial	**WARNING**
INC-FF (200 A)	FF-OFF (63 A)	3.17	14.3	1.6	Partial	**WARNING**
INC-FF (200 A)	FF-IT (40 A)	5	14.3	1.6	Partial	**WARNING**
INC-FF (200 A)	FF-LAB (80 A)	2.5	14.3	1.6	Partial	**WARNING**
INC-FF (200 A)	FF-LTG (25 A)	8	14.3	1.6	Partial	**WARNING**
INC-MCC (315 A)	MCC-WP (160 A)	1.97	19	2.5	Partial	**WARNING**
INC-MCC (315 A)	MCC-FP (125 A)	2.52	19	2.5	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y1 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R1 (10 A)	10	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y2 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B6 (10 A)	10	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y5 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B7 (10 A)	10	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R3 (40 A)	2.5	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B3 (20 A)	5	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R6 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B4 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y4 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B5 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R5 (10 A)	10	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y8 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y3 (20 A)	5	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R4 (16 A)	6.25	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B2 (6 A)	16.67	10.7	0.8	Partial	**WARNING**

Upstream	Downstream	In ratio	Fault at downstream (kA)	Selectivity limit (kA)	Short circuit	Result
INC-DBGF1 (100 A)	DB-GF1-Y7 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-B1 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-R7 (6 A)	16.67	10.7	0.8	Partial	**WARNING**
INC-DBGF1 (100 A)	DB-GF1-Y6 (16 A)	6.25	10.7	0.8	Partial	**WARNING**

8. POWER FACTOR CORRECTION

Overall status: **WARNING** (0 pass · 1 warning · 0 fail)

Method

- Strategy: Central — at the main board (MDB).
- Qc = P × (tan φ1 − tan φ2) to PF 0.95, from the maximum demand (demand factors applied). Capacitors already on the drawing are included.
- Banks are sized from the bottom up: each bank only covers what the banks below it do not, so nothing is corrected twice.
- Board banks: automatic (APFC) in 25 kvar steps, rounded up. Individual capacitors: fixed, rounded down (never over-correct a motor).
- Detuned (7 %, 189 Hz) where non-linear load ≥ 25 % of demand. Capacitor rating ≥ 1.05 × U ÷ (1 − p).
- Breaker ≥ 1.43 × bank current (IEC 60831 1.3 × overcurrent and capacitance tolerance). Light-load check at 30 % of demand.

Result

Target PF	0.95
Capacitors (new)	100 kvar in 1 banks
MDB-1 PF	0.88 → 0.954
MDB-1 capacity released	39 kVA (transformer 51 % → 47 %)

Capacitor banks

Location	Demand (kW)	Reactive (kvar)	Existing (kvar)	Banks below (kvar)	PF now	Required (kvar)	Bank	Detuning · capacitor V	Breaker · cable	PF after	Current (A)	Result
MDB-1	451	243	—	—	0.88	94.2	100 kvar (4 × 25)	None · 440 V	200 A · 4C × 70 mm²	0.954	712 → 658	**WARNING**

Main boards

Main board	Demand (kW)	PF before	PF after	kVA before → after	Released (kVA)	Transformer loading
MDB-1	451	0.88	0.954	512 → 473	39	51 % → 47 %

Notes

Location	Note
MDB-1	17 % non-linear load — check harmonics; consider a detuned bank

9. DB AND CABLE SCHEDULES

Method

- Schedules of the boards in scope, from the design data and the latest calculation run.

Result

Boards	5

DB schedule

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
MDB-1	1	INC-GF	Incomer to Ground floor SMDB	Sub-board SMDB-GF					327	500 A MCCB	TP+N	36	4C × 300 mm² XLPE/SWA + 150 mm² CPC	35	0.46	PASS	
MDB-1	2	INC-FF	Incomer to First floor SMDB	Sub-board SMDB-FF					135	200 A MCCB	TP+N	36	4C × 95 mm² XLPE/SWA + 50 mm² CPC	55	0.75	PASS	
MDB-1	3	INC-MCC	Incomer to Pump room MCC	Sub-board MCC-1					210	315 A MCCB	TP+N	36	4C × 150 mm² XLPE/SWA + 95 mm² CPC	40	0.58	PASS	
MDB-1	4	DB-EV	EV charging – parking	EV charging	44	0.7	30.8	0.98	44	63 A C	TP+N	36	4C × 10 mm² XLPE/SWA + 10 mm² CPC	60	2.38	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
MDB-1	5	DB-PV	Solar PV inverter	Solar PV	50	1	50	1	70	100 A MCCB	TP+N	36	4C × 25 mm² XLPE/SWA + 16 mm² CPC	30	0.76	PASS	
MDB-1		TOTAL	Main Distribution Board — 1600 A		514		451.1	0.88	712			29.2				45% loaded	
SMDB-GF	1	GF-LTG	Lighting	Lighting	45	0.9	40.5	0.9	63	80 A MCCB	TP+N	25	4C × 16 mm² XLPE/SWA + 16 mm² CPC	30	1.47	PASS	
SMDB-GF	2	GF-SKT	Sockets	Small power / sockets	28	0.6	16.8	0.85	27	40 A C	TP+N	25	4C × 4 mm² XLPE/SWA + 4 mm² CPC	35	2.37	PASS	
SMDB-GF	3	GF-HVAC	HVAC	HVAC	120	0.9	108	0.85	177	250 A MCCB	TP+N	25	4C × 95 mm² XLPE/SWA + 50 mm² CPC	40	1.16	PASS	
SMDB-GF	4	INC-DBGF1	Incomer to Villa ground floor DB	Sub-board DB-GF1					60	100 A MCCB	TP+N	25	4C × 35 mm² XLPE/SWA + 16 mm² CPC	30	0.92	PASS	
SMDB-GF		TOTAL	Ground floor SMDB — 630 A		231		203.3	0.87	325			21.8				52% loaded	
DB-GF1	1	DB-GF1-Y1	Majlis lighting	Lighting	1	1	1	0.9	5	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	18	1.48	PASS	
DB-GF1	2	DB-GF1-R1	Living lighting	Lighting	1.2	1	1.2	0.9	6	10 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	20	1.67	PASS	
DB-GF1	3	DB-GF1-Y2	Kitchen lighting	Lighting	0.8	1	0.8	0.9	4	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	22	1.49	PASS	
DB-GF1	4	DB-GF1-B6	Majlis sockets	Small power / sockets	1.5	1	1.5	0.9	7	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	18	1.44	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
DB-GF1	5	DB-GF1-Y5	Living sockets	Small power / sockets	2	1	2	0.9	9	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	20	1.7	PASS	
DB-GF1	6	DB-GF1-B7	Kitchen sockets	Small power / sockets	1.3	1	1.3	0.9	6	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.45	PASS	
DB-GF1	7	DB-GF1-R3	Cooker	General	6	1	6	0.9	28	40 A C	SP+N	15	2C × 6 mm² XLPE/SWA + 6 mm² CPC	24	2.79	PASS	
DB-GF1	8	DB-GF1-B3	Kitchen water heater	General	3	1	3	0.9	14	20 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	24	2.32	PASS	
DB-GF1	9	DB-GF1-R6	Maid room	Small power / sockets	1.1	1	1.1	0.9	5	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	28	1.51	PASS	
DB-GF1	10	DB-GF1-B4	Majlis split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	18	1.79	PASS	
DB-GF1	11	DB-GF1-Y4	Living split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	20	1.89	PASS	
DB-GF1	12	DB-GF1-B5	Dining split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.99	PASS	
DB-GF1	13	DB-GF1-R5	Guest bedroom	Small power / sockets	1.5	1	1.5	0.9	7	10 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	26	1.67	PASS	
DB-GF1	14	DB-GF1-Y8	Guest bathroom	Lighting	0.3	1	0.3	0.9	1	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	27	1.05	PASS	
DB-GF1	15	DB-GF1-Y3	Guest bath water heater	General	3	1	3	0.9	14	20 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	27	2.49	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
DB-GF1	16	DB-GF1-R4	Guest bedroom split A/C	HVAC	2.5	1	2.5	0.9	12	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	26	2.18	PASS	
DB-GF1	17	DB-GF1-B2	Dining lighting	Lighting	0.6	1	0.6	0.9	3	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	22	1.33	PASS	
DB-GF1	18	DB-GF1-Y7	Dining sockets	Small power / sockets	1	1	1	0.9	5	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	22	1.35	PASS	
DB-GF1	19	DB-GF1-B1	External lighting	Lighting	1	1	1	0.9	5	6 A C	SP+N	15	2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC	35	2.01	PASS	
DB-GF1	20	DB-GF1-R7	Water pump	Motor	0.8	1	0.8	0.9	3	6 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	30	1.36	PASS	
DB-GF1	21	DB-GF1-Y6	Laundry	Small power / sockets	2	1	2	0.9	9	16 A C	SP+N	15	2C × 4 mm² XLPE/SWA + 4 mm² CPC	25	1.89	PASS	
DB-GF1		TOTAL	Villa ground floor DB — 100 A		38		38	0.9	59			10.7				59% loaded	
SMDB-FF	1	FF-OFF	Office	Small power / sockets	35	0.8	28	0.9	43	63 A C	TP+N	25	4C × 10 mm² XLPE/SWA + 10 mm² CPC	25	1.66	PASS	
SMDB-FF	2	FF-IT	IT room	IT / data	20	1	20	0.95	29	40 A C	TP+N	25	4C × 4 mm² XLPE/SWA + 4 mm² CPC	20	2.04	PASS	
SMDB-FF	3	FF-LAB	Lab	General	50	0.7	35	0.85	57	80 A MCCB	TP+N	25	4C × 16 mm² XLPE/SWA + 16 mm² CPC	30	1.62	PASS	
SMDB-FF	4	FF-LTG	Corridor lighting	Lighting	4	1	4	0.9	19	25 A C	SP+N	25	2C × 6 mm² XLPE/SWA + 6 mm² CPC	35	2.57	PASS	

Board	Cct	Circuit ID	Description	Load type	Connected (kW)	DF	Demand (kW)	PF	Ib (A)	Breaker	Poles	Icu (kA)	Cable	Length (m)	Vd total (%)	Status	Accessories
SMDB-FF		TOTAL	First floor SMDB — 250 A		109		87	0.89	135			14.3				54% loaded	
MCC-1	1	MCC-WP	Water pump	Motor	75	1	75	0.86	121	160 A MCCB	TP+N	25	4C × 50 mm² XLPE/SWA + 25 mm² CPC	20	1.03	PASS	
MCC-1	2	MCC-FP	Fire pump	Fire pump	55	1	55	0.86	89	125 A MCCB	TP+N	25	4C × 35 mm² FR + 16 mm² CPC	25	1.12	PASS	
MCC-1		TOTAL	Pump room MCC — 400 A		130		130	0.86	210			19				53% loaded	

Cable schedule

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-INC-GF	MDB-1	SMDB-GF	1	Cu/XLPE/SWA/PVC (armoured)	4C × 300 mm²	150	35		327	500	525	0.46	0.46	PASS
C-INC-FF	MDB-1	SMDB-FF	6	Cu/XLPE/SWA/PVC (armoured)	4C × 95 mm²	50	55		135	200	266	0.75	0.75	PASS
C-INC-MCC	MDB-1	MCC-1	4	Cu/XLPE/SWA/PVC (armoured)	4C × 150 mm²	95	40		210	315	342	0.58	0.58	PASS
C-DB-EV	MDB-1	EV charging – parking	12	Cu/XLPE/SWA/PVC (armoured)	4C × 10 mm²	10	60		44	63	70	2.38	2.38	PASS
C-DB-PV	MDB-1	Solar PV inverter	10	Cu/XLPE/SWA/PVC (armoured)	4C × 25 mm²	16	30		70	100	120	0.76	0.76	PASS
C-GF-LTG	SMDB-GF	Lighting	11	Cu/XLPE/SWA/PVC (armoured)	4C × 16 mm²	16	30		63	80	91	1	1.47	PASS
C-GF-SKT	SMDB-GF	Sockets	14	Cu/XLPE/SWA/PVC (armoured)	4C × 4 mm²	4	35		27	40	41	1.91	2.37	PASS
C-GF-HVAC	SMDB-GF	HVAC	6	Cu/XLPE/SWA/PVC (armoured)	4C × 95 mm²	50	40		177	250	266	0.7	1.16	PASS
C-INC-DBGF1	SMDB-GF	DB-GF1	9	Cu/XLPE/SWA/PVC (armoured)	4C × 35 mm²	16	30		60	100	146	0.45	0.92	PASS

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-DB-GF1-Y1	DB-GF1	Majlis lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	18		5	6	31	0.56	1.48	PASS
C-DB-GF1-R1	DB-GF1	Living lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	20		6	10	31	0.75	1.67	PASS
C-DB-GF1-Y2	DB-GF1	Kitchen lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	22		4	6	31	0.58	1.49	PASS
C-DB-GF1-B6	DB-GF1	Majlis sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	18		7	10	41	0.52	1.44	PASS
C-DB-GF1-Y5	DB-GF1	Living sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	20		9	16	41	0.78	1.7	PASS
C-DB-GF1-B7	DB-GF1	Kitchen sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		6	10	41	0.53	1.45	PASS
C-DB-GF1-R3	DB-GF1	Cooker	25	Cu/XLPE/SWA/PVC (armoured)	2C × 6 mm²	6	24		28	40	52	1.88	2.79	PASS
C-DB-GF1-B3	DB-GF1	Kitchen water heater	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	24		14	20	41	1.4	2.32	PASS
C-DB-GF1-R6	DB-GF1	Maid room	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	28		5	6	41	0.59	1.51	PASS
C-DB-GF1-B4	DB-GF1	Majlis split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	18		12	16	41	0.87	1.79	PASS
C-DB-GF1-Y4	DB-GF1	Living split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	20		12	16	41	0.97	1.89	PASS
C-DB-GF1-B5	DB-GF1	Dining split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		12	16	41	1.07	1.99	PASS
C-DB-GF1-R5	DB-GF1	Guest bedroom	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	26		7	10	41	0.75	1.67	PASS
C-DB-GF1-Y8	DB-GF1	Guest bathroom	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	27		1	6	41	0.14	1.05	PASS
C-DB-GF1-Y3	DB-GF1	Guest bath water heater	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	27		14	20	41	1.57	2.49	PASS
C-DB-GF1-R4	DB-GF1	Guest bedroom split A/C	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	26		12	16	41	1.26	2.18	PASS

Cable tag	From	To	Ref	Type	Cores × size	CPC (mm²)	Length (m)	Tray route	Ib (A)	Breaker In (A)	Iz (A)	Vd cable (%)	Vd total (%)	Status
C-DB-GF1-B2	DB-GF1	Dining lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	22		3	6	31	0.41	1.33	PASS
C-DB-GF1-Y7	DB-GF1	Dining sockets	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	22		5	6	41	0.43	1.35	PASS
C-DB-GF1-B1	DB-GF1	External lighting	28	Cu/XLPE/SWA/PVC (armoured)	2C × 2.5 mm²	2.5	35		5	6	31	1.09	2.01	PASS
C-DB-GF1-R7	DB-GF1	Water pump	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	30		3	6	41	0.44	1.36	PASS
C-DB-GF1-Y6	DB-GF1	Laundry	26	Cu/XLPE/SWA/PVC (armoured)	2C × 4 mm²	4	25		9	16	41	0.97	1.89	PASS
C-FF-OFF	SMDB-FF	Office	12	Cu/XLPE/SWA/PVC (armoured)	4C × 10 mm²	10	25		43	63	70	0.91	1.66	PASS
C-FF-IT	SMDB-FF	IT room	14	Cu/XLPE/SWA/PVC (armoured)	4C × 4 mm²	4	20		29	40	41	1.29	2.04	PASS
C-FF-LAB	SMDB-FF	Lab	11	Cu/XLPE/SWA/PVC (armoured)	4C × 16 mm²	16	30		57	80	91	0.87	1.62	PASS
C-FF-LTG	SMDB-FF	Corridor lighting	25	Cu/XLPE/SWA/PVC (armoured)	2C × 6 mm²	6	35		19	25	52	1.82	2.57	PASS
C-MCC-WP	MCC-1	Water pump	8	Cu/XLPE/SWA/PVC (armoured)	4C × 50 mm²	25	20		121	160	174	0.45	1.03	PASS
C-MCC-FP	MCC-1	Fire pump	9 *	Fire-rated, BS 8491 (120 min, large cables)	4C × 35 mm²	16	25		89	125	146	0.54	1.12	PASS

10. ASSUMPTIONS AND LIMITATIONS

- Upstream MV network treated as infinite; transformer %Z split into R and X by its X/R ratio (default 5).
- Fault levels follow IEC 60909-0 in a simplified form, not a full study: maximum currents for breaking capacity with c_{max} = 1.05 and cable R at 20 °C; minimum currents for earth-fault disconnection with c_{min} = 0.95 and cable R at operating temperature; transformer correction K_T not applied; loads, motor contribution and generation are neglected.
- Cable ratings: the rating for each circuit's IEC 60364-5-52 installation method (E free air / tray unless set), derated for air ambient or, for buried cables, ground temperature and soil thermal resistivity, and for grouping (the cable tray route for method E, else the circuit's own parallel runs). A circuit without a rating for its method is DATA REQUIRED.
- Cable resistance at operating temperature = 1.2 × R20 (IEC 60228); protective conductor per IEC 60364-5-54 Table 54.2 unless specified. Maximum fault levels use R at 20 °C.
- Selectivity is assessed from current thresholds only; confirm with the breaker manufacturer's selectivity tables.
- Busbar voltages are measured from the main board busbar and exclude transformer regulation.

Generated by LV Design Studio 1.40.0. Calculated checks against the project limits — engineering review and approval are separate. Results must be checked by a qualified engineer before use. **Cable data — 2.5, 4, 6, 10, 16, 25, 35, 50, 95, 150, 300 mm²: app reference value (approximate, not manufacturer-verified). Verify against the manufacturer's datasheet before relying on these results.**

CALCULATION RECORD

What these results were calculated from, with what, and how far they can be relied on.

Project	Villa Complex 415V LV Network Project ID not assigned (save the project) · file format version 1
Calculated	At the time of this print, from the current inputs
Software	LV Design Studio 1.40.0 · built-in calculation engine (no external engine used)
Calculation basis	Methods per Codes and standards → Calculation basis and each study's Method notes. Standard editions: not verified against a specific edition
Key assumptions	• 415 V, 3-phase + N, 50 Hz, TN-S · design ambient 45 °C · voltage drop limit 4 % source to load • Cable resistance at operating temperature 1.2 × R20 (not set for the project) • Fault levels: maximum Ik″ with c max = 1.05 and cable R at 20 °C, MV network infinite, transformer correction KT not applied — **simplified, preliminary, not a full IEC 60909 study**
Cable data	**11 sizes used: 11 app reference values (approximate, not manufacturer-verified) · 0 manufacturer data** Final-design requirement for manufacturer cable data: not set
Input checks	No invalid inputs, no supply loops, no missing references
Calculation warnings	0 failures · 32 to check (see the results summary on page 1)
Limitations	• Cable ratings: per installation method with ambient / ground, soil and grouping factors; grouping with other circuits is counted only on cable tray routes • Selectivity from current thresholds only — confirm with the manufacturer's tables • Voltage drop: first-order, summed along the radial path; not an unbalanced load flow

Validation status: PRELIMINARY. Calculated checks against the project limits, using approximate reference or unverified cable data. Not an engineering approval and not an approval by DEWA or any other authority.